

pyCollect User Manual (GUI)

Version: 1.0

Date: 2

1. Purpose

pyCollect is a bedside monitor data collection and review application.

This manual covers the complete GUI workflow:

1. Install from setup executable.

2. Sta

This document does not cover command-line (CLI) operation.

2. System Requirements

- Windows 10 or Windows 11

- Acc

Recommended:

- Local write access to %LOCALAPPDATA%\pyCollect

- Loc

3. Installation

3.1 Install Using Setup EXE

1. Locate pyCollect_Setup.exe.

2. Ru

3.2 Installed Components

The installer places application and runtime files in these locations:

- Application: %ProgramFiles%\pyCollect (or selected install folder)

3.3 Uninstall

Uninstall from Windows Apps/Programs as usual.

Note: local runtime data under %LOCALAPPDATA%\pyCollect may be removed by uninstall policy.

4. Launching pyCollect (GUI)

Launch methods:

1. Desktop shortcut: pyCollect

2. Sta

On launch, the main window title is pyCollect Interactive Viewer.

5. Main Window Layout

The GUI is organized into a left workflow sidebar and right graph area.

5.1 Capture Tab

The CAPTURE tab contains all controls for configuring and running a live recording session.

5.2 Review Tab

The REVIEW tab provides file review, CSV conversion, and capture log summary.

5.3 Sidebar Sections

The sidebar uses a tabbed interface with CAPTURE and REVIEW tabs:

CAPTURE tab:

- STA

REVIEW tab:

- REV

Below tabs (always visible):

- Scre

5.4 Graph/Status Area

- Live trend and waveform plots

- Hea

6. Quick Start (Typical Collection Session)

1. Launch pyCollect.

2. In

7. Data Collection Procedure (Detailed)

7.1 Configure Connection

1. Open Monitor Connection.

2. Cli

7.2 Configure Output File

1. Open File Save Status.

2. Se

Behavior note:

- If the target filename already exists, pyCollect automatically switches to a timestamped non-overwriting filename and logs this action.

7.3 Set Session Duration and Display Windows

1. Open Session Setup.

2. Se

7.4 Select Channels

1. Open Waveform Selection to choose waveform channels.

2. Op

7.5 Start Recording

1. Click START.

2. Ve

7.6 Stop Recording

1. Click STOP (or let duration complete).

2. Wa

8. CSV Conversion Procedure (GUI)

After a recording is complete:

1. Go to File Save Status.

2. Cli

Expected outputs (same folder as the DRC file):

- *_trends.csv (always generated)

- *_w

9. Review Procedure (GUI)

Review mode replays the current recorded DRC file inside the GUI.

1. Ensure capture is not running.

2. En

Review mode notes:

- REVIEW operates on the current DRC file tracked in File Save Status.

- Rev

10. Color/Status Behavior (Operator Reference)

10.1 Main Capture Button

- Blue START: idle/ready

- Gre

10.2 File Save State (Current DRC File)

- Blue: appending/writing during capture

- Gre

10.3 Waveform Selection Status (catalog colors)

In simulation-oriented sessions:

- Green: selected and receiving data

- Blue

In other sessions, delayed/missing states may display warning/alarm colors.

11. Runtime Configuration Persistence

pyCollect persists runtime settings in local config so next launch can reuse prior values.

Persisted examples include:

- Selected baud rate

- Dur

Primary runtime config location:

- %LOCALAPPDATA%\pyCollect\pycollect_gui_config.json

12. Logs and Output Files

12.1 Recording Output

- DRC file: as configured in File Save Status

- Cap

12.2 GUI Status Log

- Visible in Recorder Output panel during runtime

12.3 Startup Log

- output\pycollect_qt_gui_startup.log in application/runtime working context

13. Troubleshooting

13.1 No COM Ports Visible

- Click Refresh Ports.

- Veri

13.2 Cannot Start Capture

- Confirm a COM port is selected.

- Con

If scanning shows green-highlighted source+baud combinations, prefer those first.

13.3 Flat or Missing Waveforms

- Verify correct source and baud rate.

- Che

Note: a zero-waveform capture configuration is valid; trend/alarm capture can run without waveform rows.

13.4 CSV Conversion Disabled

Conversion is enabled only when:

- capture has completed,

- curr

13.5 REVIEW Disabled

Review is enabled only when:

- capture is stopped,

- curr

14. Operational Notes

- Use consistent naming for each recording session (patient/case/date policy as defined by your site).

- Veri

15. Clinical Use Disclaimer

For research/engineering workflow support only unless explicitly validated and approved by your organization for clinical use.